

Taylor-series expansion of density functionals

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A Taylor-series expansion of the density functional $F[n]$ is constructed from the given expansion for the energy $E[v]$. The formalism is used to investigate the noninteracting kinetic-energy functional for one-dimensional systems. Furthermore, a formal analysis of the Taylor expansion of the density functional for an interacting electron system at finite temperature is given and the relation between density-functional theory and the field-theoretical approach to the many-particle problem is enlightened.

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I. INTRODUCTION

A fundamental problem in the quantum mechanics of a many-electron system can be cast into the following form: Given an external local potential $v(\mathbf{r})$, what is the corresponding ground-state energy? The answer to this question is obviously given by a functional $E[v]$. We know several routes to bring this functional to an explicit form. The representation of $E[v]$ underlying the considerations in the present paper is obtained by expanding $E[v]$ in a Taylor series

$$E[v] = E[v_0] + \int d^3r \frac{\delta E[v_0]}{\delta v(\mathbf{r})} \Delta v(\mathbf{r}) + \frac{1}{2!} \int d^3r' d^3r \frac{\delta^2 E[v_0]}{\delta v(\mathbf{r}') \delta v(\mathbf{r})} \Delta v(\mathbf{r}') \Delta v(\mathbf{r}) + \dots \quad (1)$$

The expansion coefficients in this series, i.e., the n th-order static polarization propagators, will be expressed in terms of Feynman diagrams. To obtain the polarization propagator at the correlated level, we have to add an infinite series of diagrams containing interaction vertices to the static polarization propagators calculated at the effective one-particle level.

In the density-functional approach [1] an indirect route is taken to obtain $E[v]$. The kinetic plus interaction energy $F = T + V_{ee}$ as a functional of the electron density $F = F[n]$ represents the fundamental quantity therein. In practical applications of density-functional theory one usually employs the Kohn-Sham [2] scheme, where the noninteracting kinetic-energy part $T_s[n]$ is dealt with by conventional quantum-mechanical methods and the remaining exchange correlation functional $E_{xc}[n]$ is approximated. *Ansätze* for the functional $E_{xc}[n]$ are designed in order to meet a number of known properties of the exact $F[n]$, an approach which has proven to be rather successful in practical applications (for recent reviews see, e.g., [3–7]).

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As far as systematical approaches are concerned, the methods to cast $F[n]$ into an explicit form are much less developed as in the case of $E[v]$. Hohenberg and Kohn (HK) already derived a method [1] to convert the series expansion (1) into the gradient expansion of the functional $F[n]$. However, a straightforward evaluation of the gradient corrections to the local-density approximation (LDA) does not lead to an improvement of the LDA (see, e.g., Langreth and Vosko in [6]). Important physical properties of the density functional can often only be described by infinite summations of the gradient expansion. Cures to overcome this drawback of the gradient expansion are, for instance, the generalized gradient approximation [8–11]. Furthermore, HK were able to establish a relation between the integral kernel $K(\mathbf{r} - \mathbf{r}')$

$$K(\mathbf{r} - \mathbf{r}') = \frac{\delta^2 F[n]}{\delta n(\mathbf{r}) \delta n(\mathbf{r}')} - \frac{1}{|\mathbf{r} - \mathbf{r}'|} \quad (2)$$

and the electric polarizability of the homogeneous electron gas [1]. In the present paper we will extend the latter *Ansatz* and construct the Taylor-series expansion of $F[n]$. The expansion coefficients in this series, e.g., the density response functions, will be expressed in terms of the static polarizabilities appearing in Eq. (1). The key point for this interconversion of the series expansions will be the system of equations

$$E[v] - \int d^3r n(\mathbf{r}) v(\mathbf{r}) = F[n], \quad (3)$$

$$n(\mathbf{r}) = \frac{\delta E[v]}{\delta v(\mathbf{r})}, \quad (4)$$

$$v(\mathbf{r}) = -\frac{\delta F[n]}{\delta n(\mathbf{r})}, \quad (5)$$

which shows that $F[n]$ and $E[v]$ are Legendre transforms of each other [12]. The construction of an unknown functional from the known Taylor-series expansion of its Legendre transformed functional appeared for the first time in a paper by Jona-Lasinio [13]. In order to explain the main idea in the latter scheme, let us consider Eq. (4). To calculate the derivative of both sides with respect to $n(\mathbf{r}_2)$ we make use of the chain rule

$$\begin{aligned} \frac{\delta}{\delta n(\mathbf{r}_2)} &= \int d^3 r' \frac{\delta v(\mathbf{r}')}{\delta n(\mathbf{r}_2)} \frac{\delta}{\delta v(\mathbf{r}')} \\ &= - \int d^3 r' \frac{\delta^2 F[n]}{\delta n(\mathbf{r}_2) \delta n(\mathbf{r}')} \frac{\delta}{\delta v(\mathbf{r}')} \end{aligned} \quad (6)$$

to obtain

$$\delta(\mathbf{r}_2 - \mathbf{r}) = - \int d^3 r' \frac{\delta^2 F[n]}{\delta n(\mathbf{r}_2) \delta n(\mathbf{r}')} \frac{\delta^2 E[v]}{\delta v(\mathbf{r}') \delta v(\mathbf{r})} \quad (7)$$

or, equivalently,

$$\left(\frac{\delta^2 E[v]}{\delta v(\mathbf{r}_2) \delta v(\mathbf{r}_1)} \right)^{-1} = - \frac{\delta^2 F[n]}{\delta n(\mathbf{r}_2) \delta n(\mathbf{r}_1)}. \quad (8)$$

The latter equation shows how to calculate the second-order density response function in terms of the second-order static polarization propagator. Similar relations can be derived for the higher-order density response functions and it turns out that in order to calculate the m th-order density response function we need the second-order through the m th-order static polarization propagators. The general construction scheme is given in the Appendix in a form appropriate for the considerations given here. The formal Taylor-series expansion of density functionals in terms of the polarization propagators of the homogeneous electron gas has already been discussed by Osaka [14–16]. The present work is an extension of that work in that the general structure of the expansion is investigated, not only the lowest-order terms.

The explicit representation of the density functional to be discussed here allows us to elucidate the relationship between the density-functional theory and the field-theoretical approach to the many-particle problem. Unlike in the representation of the density functional provided by the adiabatic connection approach [17–21], or the one given by Langreth [22], no integration over the coupling constant appears. The functional is solely expressed in terms of quantities which can be calculated by conventional perturbation theoretical techniques. The representation of the exchange correlation functional given by Sham and Schlüter [23–25] also avoids the coupling-constant integration, but the functional dependence on the electron density is not explicitly stated.

In addition to the formal insight into the structure of the Taylor-series expansion of the density functional, the evaluation of lower-order expansion coefficients (i.e., density response functions) and comparison with the corresponding coefficients obtained from approximate density functionals allow it to uncover deficiencies in the latter functionals. Work along this direction has already been initiated by Kugler [26]. He derived a Taylor-series expansion of the kinetic energy functional. In the present paper the work of Kugler is extended and the non-interacting kinetic-energy functional up to fourth order for a one-dimensional system is explicitly constructed.

The calculation of the Taylor-series expansion of the density functional in more complicated cases where electron-electron interaction is included relies, of course, on the accessibility of the static polarization propagators appearing in Eq. (1). In case where the infinite three-

dimensional homogeneous electron gas is used as the reference system around which the expansion Eq. (1) is made, a number of efforts have been made to calculate the second-order polarization propagator. For the noninteracting case this has been done by Lindhard [27]. The earliest attempt to include electron correlation is due to Ma and Brueckner [28]. Sham was the first one to consider the exchange contributions to the polarization propagator. Since then the calculation of the lowest-order (i.e., e^2) exchange correction to the noninteracting polarization propagator is still a matter of current research work (see [29] and references cited therein). Calculations of the linear-response function which include exchange and correlation have been reported, for instance, by Geldart and Rasolt [30] and Langreth and Perdew [31]. An overview review of the developments in this field may be found in [32,33,3].

The knowledge of the second-order polarization propagator allows a determination of the gradient coefficient in the density-gradient expansion. Furthermore, the second-order polarization propagator can be used directly in the approximation to the density functional

$$\begin{aligned} F[n] &\approx \int d^3 r g_0(n(\mathbf{r})) + \frac{1}{2} \int d^3 r d^3 r' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \\ &\quad - \frac{1}{2} \int d^2 r \int d^3 r' K[\mathbf{r} - \mathbf{r}'; n(1/2\mathbf{r} + 1/2\mathbf{r}')] \\ &\quad \times \{n(\mathbf{r}) - n(\mathbf{r}')\}^2, \end{aligned} \quad (9)$$

already proposed by HK. $g_0(n(\mathbf{r}))$ denotes the kinetic plus nonclassical electron-electron interaction-energy density of a homogeneous system, $n(1/2\mathbf{r} + 1/2\mathbf{r}')$ replaces the reference density around which the Taylor series is expanded. In a truly inhomogeneous system, however, this replacement is by no means unique and furthermore the various choices of the density argument have a non-negligible influence on the outcome of the applications of Eq. (9) [34–36].

In summary it turns out that even the calculation of the second-order polarization propagator for the homogeneous electron gas and its application provides a number of difficult problems, which are still open to further discussions. At first sight it might therefore seem that, beyond the formal achievement of bringing the Taylor-series expansion of the density functional into an explicit form, no further use could be made of this expansion. However, applications of density-functional theory deal with finite systems such as molecules or fractions of bulk material with periodic boundary conditions. For finite systems, i.e., molecules, algorithms for the calculation of the static polarization propagators of the Hartree-Fock energy up to arbitrary high order have been implemented [37]. At the correlated, multiconfigurational self-consistent-field level the calculation of the third-order response functions has been carried out [38–40] as well as for the coupled-cluster formalism [41]. Furthermore, the second-order response function has already been calculated for full configuration interaction wave functions [42]. The strong interest in the calculation of polarization propagators for finite electron systems is due to the relevance of these quantities in nonlinear optics [40].

II. AN EXPLICIT EXAMPLE FOR THE CONSTRUCTION OF THE TAYLOR-SERIES EXPANSION OF A DENSITY FUNCTIONAL

In this section we want to give a simple example for the construction of the Taylor-series expansion of a density functional. The construction of any approximation to the density functional relies on the knowledge of properties such as scaling properties and sum rules obeyed by the exact density functional and the construction scheme

discussed in the present paper offers a systematic tool for uncovering the properties of the exact density functional.

To fix the ideas we choose a many-electron system at zero temperature, which is described in terms of a one-particle Hamiltonian

$$\hat{h} = \hat{T} + \hat{v}_{nl} + \hat{v}_0, \quad (10)$$

where $\hat{v}_0$ denotes a local and $\hat{v}_{nl}$ a nonlocal one-particle potential. The expansion of the corresponding density functional is then given by (see the Appendix)

$$F[n] = F[n_0] + \text{diagram 1} - \frac{1}{2!} \text{diagram 2} + \frac{1}{3!} \text{diagram 3} - \frac{3}{4!} \text{diagram 4} + \frac{1}{4!} \text{diagram 5} + \dots, \quad (11)$$

The diagrams are as follows:

- Diagram 1: A circle labeled F connected to a circle labeled Δn .
- Diagram 2: Two circles labeled Δn connected by a horizontal line.
- Diagram 3: A circle labeled E connected to three circles labeled Δn (one above, one below, and one to the right).
- Diagram 4: Two circles labeled E connected by a horizontal line, with two circles labeled Δn above each E and two circles labeled Δn below each E .
- Diagram 5: A circle labeled E connected to three circles labeled Δn (one above, one below, and one to the right), with an additional circle labeled Δn to the left.

where

$$\begin{aligned} \text{Diagram 1} &= \frac{\delta F[n_0]}{\delta n(\mathbf{r})} \\ &= -v(\mathbf{r}), \\ \text{Diagram 3} &= \frac{\delta^n}{\delta v(\mathbf{r}_n) \dots \delta v(\mathbf{r}_1)} E[v], \end{aligned} \quad (12)$$

and the dot $\bullet$ denotes the inverse of the kernel $\frac{\delta^2}{\delta v(\mathbf{r}_2) \delta v(\mathbf{r}_1)} E[v]$.

In order to calculate the static polarization propagators we consider the expansion of the ground-state energy in terms of a perturbing external local potential $\Delta v(\mathbf{r})$. In the case where we use the Hamiltonian Eq. (10) this expansion is just the familiar perturbation expansion. With the conventional diagrammatic technique [43] one obtains, up to fourth order,

$$\begin{aligned} E = & E[v_0] \\ & + \sum_{hh'} n_{hh'} \Delta v_{hh'} \\ & + \sum_{hp} \frac{\Delta v_{hp} \Delta v_{ph}}{(\epsilon_h - \epsilon_p)} \\ & + \sum_{hpp'} \frac{\Delta v_{hp} \Delta v_{pp'} \Delta v_{p'h}}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'})} - \sum_{hh'p} \frac{\Delta v_{hp} \Delta v_{ph'} \Delta v_{h'h}}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)} \\ & + \sum_{hpp'p''} \frac{\Delta v_{hp} \Delta v_{pp'} \Delta v_{p'p''} \Delta v_{p''h}}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'}) (\epsilon_h - \epsilon_{p''})} + \sum_{hh'h''p} \frac{\Delta v_{hp} \Delta v_{ph''} \Delta v_{h''h'} \Delta v_{h'h}}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)(\epsilon_{h''} - \epsilon_p)} \\ & - \sum_{hh'pp'} \frac{\Delta v_{hp} \Delta v_{pp'} \Delta v_{p'h'} \Delta v_{h'h}}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'}) (\epsilon_{h'} - \epsilon_{p'})} - \sum_{hh'pp'} \frac{\Delta v_{hp} \Delta v_{pp'} \Delta v_{p'h'} \Delta v_{h'h}}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)(\epsilon_{h'} - \epsilon_{p'})} \\ & - \sum_{hh'pp'} \frac{\Delta v_{hp} \Delta v_{ph'} \Delta v_{h'p'} \Delta v_{p'h}}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'}) (\epsilon_{h'} - \epsilon_{p'})} \\ & + \dots, \end{aligned} \quad (13)$$

where p refers to particle and h to hole states. The second derivative of E with respect to the external local potential $\Delta v(\mathbf{r})$, i.e., the static polarization propagator $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ as obtained from Eq. (13), is given by

$$\begin{aligned}\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0) &= \sum_{p'h'ph} \frac{\delta \Delta v_{h'p'}}{\delta \Delta v(\mathbf{r}')} \frac{\delta^2 E[v_0]}{\delta \Delta v_{h'p'} \delta \Delta v_{ph}} \frac{\delta \Delta v_{ph}}{\delta \Delta v(\mathbf{r})} + \text{c.c.} \\ &= \sum_{ph} \frac{2}{(\epsilon_h - \epsilon_p)} \text{Re} \varphi_h^*(\mathbf{r}') \varphi_p(\mathbf{r}') \varphi_p^*(\mathbf{r}) \varphi_h(\mathbf{r}),\end{aligned}\quad (14)$$

where c.c. stands for complex conjugate. For notational convenience we define an additional local potential by

$$\begin{aligned}u(\mathbf{r}) &= \text{---} \bigcirc \Delta n \\ &= \int d^3 r' \mathcal{D}^{-1}(\mathbf{r}, \mathbf{r}', \omega = 0) \Delta n(\mathbf{r}').\end{aligned}\quad (15)$$

The expansion of $F[n] = \langle \psi_0[n] | \hat{T} + \hat{v}_{nl} + \hat{v}_0 | \psi_0[n] \rangle$, where $\psi_0[n]$ denotes the ground-state wave function, in terms of the density becomes

$$\begin{aligned}F[n] &= F[n_0] \\ &- \int d^3 r v_0(\mathbf{r}) \Delta n(\mathbf{r}) \\ &- \frac{1}{2} \int d^3 r u(\mathbf{r}) \Delta n(\mathbf{r}) \\ &+ \int d^3 r_1 d^3 r_2 d^3 r_3 u(\mathbf{r}_1) u(\mathbf{r}_2) u(\mathbf{r}_3) \\ &\times \left\{ \sum_{h p p'} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{pp'}(\mathbf{r}_2) \varphi_{p'h}(\mathbf{r}_3)}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'})} - \sum_{h h' p} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{ph'}(\mathbf{r}_2) \varphi_{h'h}(\mathbf{r}_3)}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)} \right\} \\ &- \int d^3 r_1 d^3 r_2 d^3 r_3 d^3 r_4 u(\mathbf{r}_1) u(\mathbf{r}_2) u(\mathbf{r}_3) u(\mathbf{r}_4) \int d^3 r' d^3 r'' \\ &\times \frac{3}{4!} \left\{ \sum_{h p p'} \frac{\varphi_{pp'}(\mathbf{r}_1) \varphi_{p'h}(\mathbf{r}_2) \varphi_{hp}(\mathbf{r}')}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'})} - \sum_{h'} \frac{\varphi_{ph'}(\mathbf{r}_1) \varphi_{h'h}(\mathbf{r}_2) \varphi_{hp}(\mathbf{r}')}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)} \right\} \mathcal{D}^{-1}(\mathbf{r}', \mathbf{r}'', \omega = 0) \\ &\times \left\{ \sum_{\bar{h} \bar{p} \bar{p}'} \frac{\varphi_{\bar{p}\bar{p}'}(\mathbf{r}_3) \varphi_{\bar{p}'\bar{h}}(\mathbf{r}_4) \varphi_{\bar{h}\bar{p}}(\mathbf{r}'')}{(\epsilon_{\bar{h}} - \epsilon_{\bar{p}})(\epsilon_{\bar{h}} - \epsilon_{\bar{p}'})} - \sum_{h'} \frac{\varphi_{\bar{p}\bar{h}'}(\mathbf{r}) \varphi_{\bar{h}'\bar{h}}(\mathbf{r}) \varphi_{\bar{h}\bar{p}}(\mathbf{r}'')}{(\epsilon_{\bar{h}} - \epsilon_{\bar{p}})(\epsilon_{\bar{h}'} - \epsilon_{\bar{p}})} \right\} \\ &- \text{(terms obtained by permuting } \mathbf{r}_1, \mathbf{r}_2, \mathbf{r}' \text{ and } \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}'', \text{ respectively)} \\ &+ \int d^3 r_1 d^3 r_2 d^3 r_3 d^3 r_4 u(\mathbf{r}_1) u(\mathbf{r}_2) u(\mathbf{r}_3) u(\mathbf{r}_4) \\ &\times \left\{ \sum_{h p p' p''} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{pp'}(\mathbf{r}_2) \varphi_{p'p''}(\mathbf{r}_3) \varphi_{p''h}(\mathbf{r}_4)}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'})(\epsilon_h - \epsilon_{p''})} + \sum_{h h' h'' p} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{ph''}(\mathbf{r}_2) \varphi_{h''h'}(\mathbf{r}_3) \varphi_{h'h}(\mathbf{r}_4)}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)(\epsilon_{h''} - \epsilon_p)} \right. \\ &- \sum_{h h' p p'} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{pp'}(\mathbf{r}_2) \varphi_{p'h'}(\mathbf{r}_3) \varphi_{h'h}(\mathbf{r}_4)}{(\epsilon_h - \epsilon_p)(\epsilon_h - \epsilon_{p'})(\epsilon_{h'} - \epsilon_{p'})} - \sum_{h h' p p'} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{pp'}(\mathbf{r}_2) \varphi_{p'h'}(\mathbf{r}_3) \varphi_{h'h}(\mathbf{r}_4)}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_p)(\epsilon_{h'} - \epsilon_{p'})} \\ &+ \left. \sum_{h h' p p'} \frac{\varphi_{hp}(\mathbf{r}_1) \varphi_{ph'}(\mathbf{r}_2) \varphi_{h'p'}(\mathbf{r}_3) \varphi_{p'h}(\mathbf{r}_4)}{(\epsilon_h - \epsilon_p)(\epsilon_{h'} - \epsilon_{p'})(\epsilon_{h'} - \epsilon_p)} \right\} \\ &+ \dots,\end{aligned}\quad (16)$$

where $F[n_0]$ is given by $E[v_0] - \int v_0 n_0$ and $\varphi_{ij}(\mathbf{r})$ abbreviates $\varphi_i^*(\mathbf{r}) \varphi_j(\mathbf{r})$, where the spin degrees of freedom are understood to be integrated over. The calculation of higher-order terms is straightforward, but the algebra becomes more involved.

The expansion coefficients depend obviously on the zeroth-order reference point characterized by $v_0(\mathbf{r})$. In this sense the representation of the density functional

discussed here follows an indirect route via conventional quantum-mechanical and field-theoretical methods. $F[n]$ is not obtained from a first-principles density-functional theory approach, such as, for instance, a functional differential equation in $n(\mathbf{r})$. The actual construction of the Taylor series of $F[n]$ is of comparable complexity to the construction of the expansion of $E[v]$, which means that the representation of $F[n]$ discussed here is not suitable

for actual applications of density-functional theory.

In the case where only the lowest hole level is doubly occupied, the series Eq. (16) is identical to an expression of $\langle \psi_0^b[n] | \hat{T} + \hat{v}_{nl} + \hat{v}_0 | \psi_0^b[n] \rangle$, where n is the density of a two-boson system and $\psi_0^b[n]$ the corresponding ground-state wave function. For $v_{nl} = v_0 = 0$ Eq. (16) sums up to $\frac{1}{8m} \int d^3r \frac{|\nabla n(\mathbf{r})|^2}{n(\mathbf{r})}$, the kinetic energy of a boson system of density $n(\mathbf{r})$. Another less obvious example for which the later expansion of a density functional could be summed up to infinite order will be given later in this section.

To further specify Eq. (16) we consider the case where the electrons move in an external potential which is constant in the x direction and which gives rise to a small spatial extension of the electron density in the y and z directions. The latter situation is realized by the π -electron system in aromatic molecules, which are often described within the Hückel theory of aromatic molecules (for a review see [44]). In the continuous version of this model, the electrons are moving in one dimension in the absence of an external potential. The electron orbitals $\varphi_k(x)$ are subjected to periodic boundary conditions, i.e.,

$$\varphi_k(x + L) = \varphi_k(x). \quad (17)$$

The eigenvalues for this system ϵ_k become (here and in the following we will use atomic units)

$$\epsilon_k = \frac{k^2}{2}, \quad k = \frac{2\pi}{L}n, \quad n \in \mathbb{Z} \quad (18)$$

and the eigenfunctions are given by

$$\varphi_k(x) = \sqrt{\frac{1}{L}} e^{-ikx}. \quad (19)$$

The spatial orbitals $\varphi_k(x)$ and $\varphi_{-k}(x)$ are degenerate and describe electrons moving left or right around the cycle. In the following we will restrict our considerations to closed-shell systems only, e.g., systems with $2 + 4n$ electrons, since open-shell states are degenerate so that the Legendre transformation is not applicable [12]. The ground-state energy can be obtained from

$$\begin{aligned} E[v_0 = 0] &= \frac{2\pi^2}{L^2} \sum_{n=1}^{(N-2)/4} 4n^2 \\ &= \frac{4}{3} \frac{\pi^2}{L^2} M(M+1)(2M+1), \quad M = (N-2)/4. \end{aligned} \quad (20)$$

This expression is the analog to the Thomas-Fermi term [45,46] for the kinetic energy of independent particles as a function of the electron density.

In view of the large amount of details contained in the density response functions we restrict our discussion to a few qualitative aspects. The extension of our approach to the more relevant — in the sense of practical applications — three-dimensional case will be discussed elsewhere. With the preceding settings we obtain, for the

kernel Eq. (14),

$$\begin{aligned} \frac{\delta^2 E[v]}{\delta v(x') \delta v(x)} &= \frac{4}{L} \sum_{ph} \frac{1}{[-2h(p-h) - (p-h)^2]} \\ &\quad \times \varphi_{(p-h)}^*(x') \varphi_{(p-h)}(x) + \text{c.c.} \\ &= \frac{8}{L} \sum_{hq} \frac{\Theta(|h+q| - k_f)}{(-2hq - q^2)} \text{Re} \varphi_q^*(x') \varphi_q(x), \end{aligned}$$

$$q = p - h, \quad (21)$$

where c.c. is used as an abbreviation for the complex conjugate of the preceding term and Θ is defined by $\Theta(x) = 1$ for $x > 0$ and $\Theta(x) = 0$ elsewhere. An additional factor of 2 is included on the right-hand side (rhs) to account for the spin degeneracy of the electron orbitals; the indices h and p therefore refer to spatial orbitals, not to spin orbitals now. Furthermore, we have made use of the fact that $\varphi_h^*(x) \varphi_p(x) = \sqrt{\frac{1}{L}} \varphi_{h-p}^*(x)$ in deriving Eq. (21). The m th-order static polarization propagators similarly can readily be written as functions of the momenta $q_1, \dots, q_m$ transferred to the system as result of the interaction with the external potential Δv_{q_i} . Equation (21) represents the finite-system analog to the Lindhard density response function in one dimension [47]. The inverse of this function is easy to calculate since the $\varphi_{h-p}^*(x)$ form an orthogonal set of functions. We find

$$\begin{aligned} \left\{ \frac{\delta^2 E[v]}{\delta v(x') \delta v(x)} \right\}^{-1} &= \frac{L}{8} \sum_q \left\{ \sum_h \frac{\Theta(|h+q| - k_f)}{(-2hq - q^2)} \right\}^{-1} \\ &\quad \times \text{Re} \varphi_q^*(x') \varphi_q(x), \end{aligned} \quad (22)$$

where we have introduced the Fermi vector $k_f = \frac{2\pi}{L}(N/2 - 1)/2$.

The electron density will now be expanded in a Fourier series

$$\Delta n(x) = \sum_{l=-\infty}^{\infty} g_l e^{-ilx}, \quad g_0 = 0, \quad g_l = g_{-l}^* \quad (23)$$

and we turn our attention to the second-order term in the expansion Eq. (16). Since this term causes no coupling of density oscillations with different frequencies, we can restrict our discussion to electron densities $\Delta n(x)$ made up of one single oscillating term $\Delta n(x) = 1/2(e^{-il_1x} + e^{il_1x}) = \cos(l_1x)$. The curves in Fig. 1 for the same value of the electron density on cycles of different lengths reflect nicely the particle-number dependence, or in other words, the derivative discontinuity [48] of the density functional.

With an increasing number of electrons the second-order contribution approaches the value obtained with the expression [49]

$$F_2[n] = -\frac{L}{2} \sum_l \frac{|g_l|^2}{\mathcal{D}^\infty(l, \omega = 0)}, \quad (24)$$

where the Fourier components of the Lindhard density response function $\mathcal{D}^\infty(l, \omega = 0)$ for an infinite system in

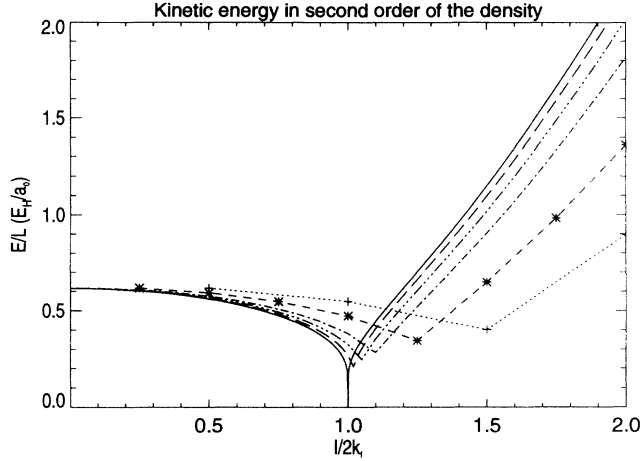


FIG. 1. The figure shows the contribution to the kinetic energy in second order of the electron density for a range of different particle numbers. The length L of the cycle is equal to the particle number and the energy is scaled with the inverse length of the system. The particle numbers corresponding to the curves from left to right are ∞ , 102, 42, 22, 10, and 6, respectively. The curve for 1002 particles would be completely covered by the solid curve. The calculated values are marked on the interpolation graphs for six and ten electrons only.

one dimension [49] are given by

$$\mathcal{D}^\infty(l, \omega = 0) = -\frac{1}{\pi k_f} \eta^{-1} \ln \left| \frac{1 + \eta}{1 - \eta} \right|, \quad \eta = |l/2k_f|. \quad (25)$$

Figure 1 shows that the second-order term in Eq. (16) favors density oscillations with a frequency of $2k_f + \frac{2\pi}{L}$ over other frequencies. However, in contrast to what is predicted by Eq. (24), the kinetic-energy functional of any finite system exhibits an energy change different from zero at $l = 2k_f + \frac{2\pi}{L}$. That means that the finite homogeneous electron system is not degenerate with the system which shows density oscillations. Furthermore, the destabilization upon density oscillations in the finite periodic system has its minimum at $l = 2k_f + \frac{2\pi}{L}$; there are as many maxima in the density as there are electron pairs for this frequency. However, in the limit of an infinite system $\frac{2\pi}{L}$ becomes zero, so that we would deduce the wrong shell structure for the finite system by looking at the infinite system.

The strong tendency in a one-dimensional system toward the formation of electron pairs has its physical explanation in the fact that an electron pair forms a boson and that in pairing themselves the electrons circumvent the Pauli exclusion principle to a certain extent. The latter effect becomes manifest in the Peierls distortion of quasi-one-dimensional systems such as polyacetylene [50].

Figure 1 indicates that density oscillations with frequency $l = 2k_f + \frac{2\pi}{L}$ are damped, in the potential $u(x)$, for instance, by the propagator $\frac{\delta^2 F[n]}{\delta n(x') \delta n(x)} =$

$-\mathcal{D}^{-1}(x', x, \omega = 0)$ of density disturbances (cf. Sec. III). Therefore a better convergence of the expansion (16) as compared to the perturbation expansion (13) is expected for Δv or Δn , respectively, in this frequency range. For rapidly oscillating densities with nonvanishing Fourier components g_q , only for $|q| \gg 2k_f$ the kernel (22) reduces to

$$\mathcal{D}^{-1}(x', x, \omega = 0) \stackrel{|q| \gg 2k_f}{=} -\frac{L}{4N} \sum_q q^2 \text{Re} \varphi_q^*(x') \varphi_q(x), \quad (26)$$

so that the second-order kinetic-energy term $F_2[n]$ becomes

$$F_2[n] = \frac{L^2}{8N} \sum_l |g_l|^2 l^2. \quad (27)$$

This is the von Weizsäcker correction [51] for a finite one-dimensional system [49] with an electron density deviating weakly from uniformity. The corresponding relation between Eqs. (26) and (27) in three dimensions has been given by Jones and Young [47]. The latter consideration provides an example where the electron system behaves like a system of bosons since formula (27) holds for bosons too. The reason for this coincidence between electronic and bosonic systems is simply the fact that in second-order perturbation theory large momenta ($|q| \gg 2k_f$) transferred to the system cannot resolve the fermionic structure; the Pauli principle does not show up because only excitations into orbitals with momenta far above the Fermi surface are possible and, furthermore, the difference in momenta of the occupied states is negligible compared to the momentum transferred to the system. However, in general, in m th-order ($m > 2$) of perturbation theory the fact that the momenta $q_1, \dots, q_n$ which are transferred to the system are large in absolute magnitude is not sufficient to ensure a bosonlike behavior of electron systems since successive transfers of large momenta could result in an overall transfer of momentum which is not large in absolute magnitude compared with the Fermi momentum. However, if we additionally require that any possible combination of momenta, which are different in absolute magnitude, is again large compared with the Fermi momentum, we retain the analogy between fermionic and bosonic systems. The latter observation implies that the expansion of the kinetic-energy density functional could be summed up under the given assumptions to result in the von Weizsäcker term.

In one-dimensional kinetic-energy functionals constructed so far [49,52] the second-order energy change Eq. (24) has been used as a guide in the construction of the functionals. As a qualitative feature which is independent of the number of particles in the closed-shell system, this second-order term shows the tendency towards density oscillations. However, $\mathcal{D}^\infty$ alone gives of course no information about the particle number dependence of the kinetic-energy functional for finite systems. Furthermore, no information about the interplay between density oscillations with different frequencies can be obtained from this term.

To provide some insight into the latter problem we

turn our attention to the third- and fourth-order terms in Eq. (16). In the discussion of the information contained in these terms we restrict ourselves to an electron density Δn consisting of two oscillating terms

$$\Delta n(x) = \cos(l_1 x) + \cos(l_2 x). \quad (28)$$

Numerical results for the third-order contribution to the kinetic energy are shown in Fig. 2. The derivative discontinuity shows up again in different curves for the various particle numbers. Due to momentum conservation this term gives a nonzero contribution only if the absolute value of one frequency is twice as large as the absolute value of the other one. As a consequence, the third-order term does not give a contribution to the self-interaction of a density oscillation. The minima in the interpolation curve at $2l_2 = l_1 = 2k_f$ and $2l_2 = l_1 = 2(2k_f + \frac{2\pi}{L})$ have an obvious physical interpretation. In the former case the rapid density oscillation separates all but two of the electrons into pairs and the slower density oscillation leads to an alternation of more and less pronounced electron pairs. In the latter case the electrons are separated into pairs by the slower oscillation, while the faster one gives rise to a deformation of the individual electron pairs and therefore increases the energy.

The fourth-order term of the kinetic energy does not have a restriction on the ratio of the frequencies l_1 and l_2 . The resulting surface of a plot of $F_4[n]$ against $l_1/2k_f$ and $l_2/2k_f$ for a system containing ten electrons is given in Fig. 3. The surface for a 102-particle system is shown in Fig. 4. The qualitative structure of this surface is roughly the same as in the ten-electron case, but more pronounced and better resolved and we will therefore an-

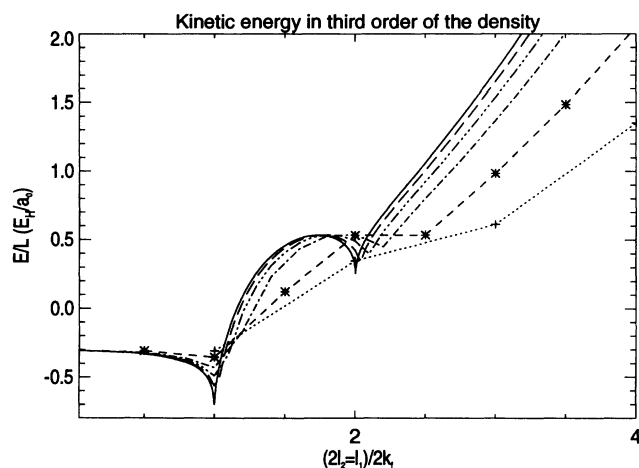


FIG. 2. Kinetic energy in third order of the electron density. As in Fig. 1 the length L of the cycle is equal to the particle number and the energy is scaled with the inverse length of the system. The nonvanishing contributions that arise for $2l_2 = l_1$ (and for l_1 interchanged with l_2) are interpolated by straight lines. The particle numbers corresponding to the curves in the upper right corner are, from left to right, 1002, 102, 42, 22, 10, and 6, respectively. Again the calculated values are marked on the interpolation graphs for six and ten electrons only.

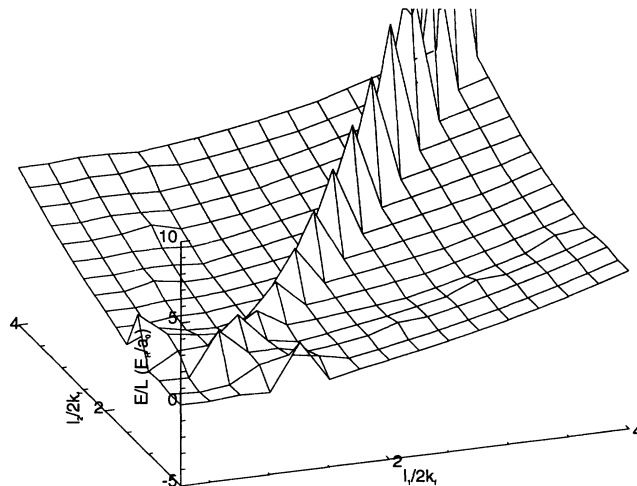


FIG. 3. Fourth-order term $F_4[n]/L$ for a system with ten electrons and a cycle length L of $L = 10$. As in Figs. 1 and 2, the z axis is scaled with the length of the cycle.

alyze the latter one. The fourth-order term obviously diminishes the relative stabilization of density oscillations with frequencies $2k_f + \frac{2\pi}{L}$. At $l_1 = l_2 = 2k_f + \frac{2\pi}{L}$ we find a very sharp δ -function-like singularity which describes the self-interaction of the two density oscillations. The above mentioned damping of density oscillations with frequency equal to $2k_f$ by $\mathcal{D}^{-1}(x, x', \omega = 0)$ is not sufficient, however, to suppress this singularity. If one of the frequencies takes the value k_f , all but two of the electrons can be separated into quadruples and we see a slight preference for this structure. For $l_1 = l_2 = k_f$ this effect becomes particularly pronounced, as can be seen in Fig. 5. Close to this point we find a valley above the $(l_1 + l_2) = 2k_f + \frac{2\pi}{L}$ axis, which is again caused by the tendency towards the formation of electron pairs. Electron pairs are transferred from one density oscillation to the other while moving around in this valley.

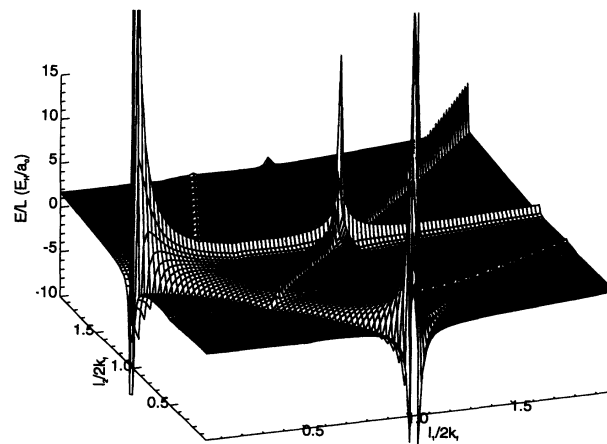


FIG. 4. The term $F_4[n]/L$ for a system with 102 electrons and a cycle length of 102.

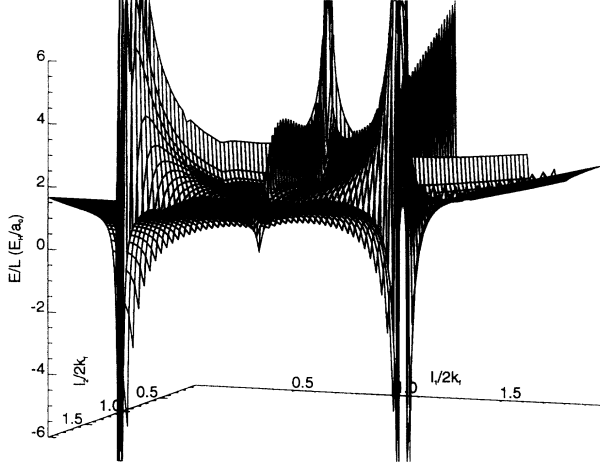


FIG. 5. The graph of Fig. 4, shown from below.

A further interesting structure can be found above the parallels to the $l_1 = l_2$ axis near the points $l_1 = 2k_f + \frac{4\pi}{L}$, $l_2 = \frac{2\pi}{L}$ and $l_2 = 2k_f + \frac{4\pi}{L}$, $l_1 = \frac{2\pi}{L}$, respectively. Away from these points the observed relative stabilization vanishes rapidly. The latter structure has no obvious explanation in terms of the formation of electron pairs. However, it is easy to see that the above specified combination of momenta allows it to couple occupied with unoccupied states, which are both close to the Fermi surface.

In concluding this section we note that the basic qualitative features as well as quantitative information about the particle number dependence, for instance, are accessible via the given expansion of the kinetic-energy functional.

III. FORMAL PROPERTIES OF THE TAYLOR-SERIES EXPANSION OF THE INTERACTING DENSITY FUNCTIONAL AT FINITE TEMPERATURE

After having investigated a particular example for the Taylor-series expansion of a density functional we will now discuss the formal aspects of the Taylor-series expansion of the density functional for interacting electrons. For the sake of generality an electron system at finite temperature [53] will be considered. The finite-temperature generalization $F_\beta[n]$ of $F[n]$ can be generated from the grand canonical potential $\Omega[\alpha](\beta)$ (for the definition of $F_\beta[n]$ and $\Omega[\alpha](\beta)$ see [12]) by means of a Legendre transformation [12]. For an alternative to the derivation of the finite-temperature generalization of $F[n]$ given in [12], we refer to the book of Parr and Yang [4]. These authors developed a finite-temperature density-functional theory within the constrained search approach of Levy and Lieb [54–57].

We define $v'(\mathbf{r}) = -\frac{1}{\beta}\alpha(\mathbf{r})$ such that the density becomes the quantity conjugate to $v'(\mathbf{r})$ with respect to $\Omega[-\beta v'](\beta)$, i.e., $n(\mathbf{r}) = \frac{\delta\Omega[-\beta v'](\beta)}{\delta v'(\mathbf{r})}$. The Taylor-series expansion of $F_\beta[n]$ can now be obtained by simply replacing the static polarization propagators corresponding to

$E[v]$ in Eq. (11) by the finite-temperature polarization propagators of the grand canonical potential.

Expansions of Legendre transformed functionals such as the one under consideration are frequently used in quantum field theory [58,59] and we adopt a field-theoretical point of view in the discussion of the expansion of $F_\beta[n]$. The potential v' in $\Omega[-\beta v'](\beta)$ is considered to be the source strength of scalar bosons to which the electrons couple via $\int d^3r v'(\mathbf{r})\hat{n}(\mathbf{r})$. The variable conjugate to the external source strength with respect to $\Omega[-\beta v'](\beta)$ is the boson field, here given by n . The coefficient quadratic in the external source, namely, the $\omega = 0$ component of the retarded polarization propagator $D^R(\mathbf{r}', \mathbf{r}, \omega = 0)$

$$D^R(\mathbf{r}', \mathbf{r}, \omega = 0) = \frac{\delta^2}{\delta v'(\mathbf{r}')\delta v'(\mathbf{r})}\Omega[-\beta v'](\beta), \quad (29)$$

plays the role of the propagator of the bosons. Due to the symmetry between a functional and its Legendre transform, v' and n interchange their role by going from $\Omega[-\beta v'](\beta)$ to $F_\beta[n]$ and $-D^{R^{-1}}(\mathbf{r}', \mathbf{r}, \omega = 0)$ appears as the propagator of the scalar particle the source of which is n . It describes the spreading of a static density disturbance like $D^R(\mathbf{r}', \mathbf{r}, \omega = 0)$ describes the spreading of a static change in the external potential with respect to the reference many-particle medium. The m th-order polarizabilities Eq. (12) with $m \geq 3$ present in the expansion of the Legendre transform are the interaction vertices of the density disturbances. They are connected to each other and to the field via the propagator $D^{R^{-1}}(\mathbf{r}', \mathbf{r}, \omega = 0)$.

To circumvent the difficulties with noncommuting operators in the calculation of the propagator and the interaction vertices we assign a formal time label to the operators, so that the grand canonical potential becomes

$$\Omega[-\beta v'](\beta) = -\frac{1}{\beta} \ln \text{Tr} \mathcal{T} \exp \left[- \int_0^\beta d\tau \left(\hat{H}'(\tau) + \int d^3r v'(\mathbf{r}, \tau) \hat{n}(\mathbf{r}, \tau) \right) \right]. \quad (30)$$

$\mathcal{T}$ denotes the time-ordering operator and $\hat{H}'$ is the sum of the kinetic-energy operator plus the electron-electron interaction operator. We consider, for instance,

$$\begin{aligned} & \frac{\delta^2}{\delta v'(\mathbf{r}')\delta v'(\mathbf{r})}\Omega[-\beta v'](\beta) \\ &= -\frac{1}{\beta} \left(\int_0^\beta d\tau \int_0^\beta d\tau' \langle \mathcal{T} \hat{n}(\mathbf{r}', \tau') \hat{n}(\mathbf{r}, \tau) \rangle \right. \\ & \quad \left. - \int_0^\beta d\tau \langle \mathcal{T} \hat{n}(\mathbf{r}', \tau') \rangle \int_0^\beta d\tau' \langle \mathcal{T} \hat{n}(\mathbf{r}, \tau) \rangle \right). \end{aligned} \quad (31)$$

In order to further discuss the term on the rhs of Eq. (31) and to establish a relation to the elementary perturbation theory used by Mermin to obtain an expression for the integral kernel $K(\mathbf{r}' - \mathbf{r})$, we insert completeness relations into Eq. (31) and obtain the Lehmann representation of the imaginary-time polarization propagator

$$\begin{aligned}
\mathcal{D}(\mathbf{r}', \mathbf{r}, \tau' - \tau) &= \langle \mathcal{T} \hat{n}(\mathbf{r}', \tau') \hat{n}(\mathbf{r}, \tau) \rangle - \langle \mathcal{T} \hat{n}(\mathbf{r}', \tau') \rangle \langle \mathcal{T} \hat{n}(\mathbf{r}, \tau) \rangle \\
&= \frac{1}{Z} \sum_{i,j,N} \left[\Theta(\tau - \tau') e^{(\tau - \tau')(E_i^N - E_j^N)} + \Theta(-\tau + \tau') e^{-(\tau - \tau')(E_i^N - E_j^N)} \right] e^{-\beta E_i^N} \\
&\quad \times \langle \psi_i^N | \hat{n}(\mathbf{r}') | \psi_j^N \rangle \langle \psi_j^N | \hat{n}(\mathbf{r}) | \psi_i^N \rangle - n(\mathbf{r}') n(\mathbf{r}).
\end{aligned} \tag{32}$$

Here we have made use of the fact that the i th N -particle eigenfunction ψ_i^N of the Hamiltonian $\hat{H}' + \hat{v}'$ can be chosen to be real so that $\langle \psi_j^N | \hat{n}(\mathbf{r}) | \psi_i^N \rangle = \langle \psi_i^N | \hat{n}(\mathbf{r}) | \psi_j^N \rangle$. The double integration in Eq. (31) then leads to $-\beta$ times the term

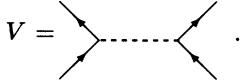
$$\begin{aligned}
&-\frac{1}{Z} \sum_{i,j,N} \frac{e^{-\beta E_j^N} - e^{-\beta E_i^N}}{E_i^N - E_j^N} \langle \psi_i^N | \hat{n}(\mathbf{r}') | \psi_j^N \rangle \\
&\quad \times \langle \psi_j^N | \hat{n}(\mathbf{r}) | \psi_i^N \rangle + \beta n(\mathbf{r}') n(\mathbf{r}), \tag{33}
\end{aligned}$$

which is equal to $D^R(\mathbf{r}', \mathbf{r}, \omega = 0)$ and to $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$. A representation similar to Eq. 33 can, of course, be obtained also for the higher-order polarization propagators.

The inverse of $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ plays an important role in the expansion of $F_\beta[n]$ and a few remarks about this function are in order. $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ can be obtained according to the Dyson equation

$$\begin{aligned}
\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0) &= -V^{-1}(\mathbf{r}' - \mathbf{r}) + \int d^3 \tilde{\mathbf{r}} d^3 \tilde{\mathbf{r}} V^{-1}(\mathbf{r}' - \tilde{\mathbf{r}}) \\
&\quad \times \mathcal{D}^*(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}, \omega = 0) \mathcal{D}(\tilde{\mathbf{r}}, \mathbf{r}, \omega = 0). \tag{34}
\end{aligned}$$

$\mathcal{D}^*(\mathbf{r}', \mathbf{r}, \omega = 0)$ denotes the irreducible part of the polarization propagator, which cannot be separated into two parts by cutting a through a single vertex line



The interaction potential V is given by $V(\mathbf{r}' - \mathbf{r}) = \frac{e^2}{|\mathbf{r}' - \mathbf{r}|}$. We use Eq. (34) for $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ instead of the conventional Dyson equation

$$\begin{aligned}
\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0) &= \mathcal{D}^*(\mathbf{r}', \mathbf{r}, \omega = 0) \\
&\quad + \int d^3 \tilde{\mathbf{r}} d^3 \tilde{\mathbf{r}} \mathcal{D}^*(\mathbf{r}', \tilde{\mathbf{r}}, \omega = 0) \\
&\quad \times V(\tilde{\mathbf{r}} - \tilde{\mathbf{r}}) \mathcal{D}(\tilde{\mathbf{r}}, \mathbf{r}, \omega = 0) \tag{35}
\end{aligned}$$

to emphasize that V^{-1} is the zeroth-order propagator of the bosons, the source of which is the external potential v' . The interaction of the bosons is mediated by the electrons in the system. $\mathcal{D}^*(\mathbf{r}', \mathbf{r}, \omega = 0)$ therefore appears as a mass operator in Eq. (34).

The situation is reversed with respect to the quantum-mechanical approach to the many-electron problem via the Schrödinger equation. In the latter case the electronic degrees of freedom are treated explicitly, while the bosons appear via their inverse propagator V as an electron-electron interaction.

From Eq. (35) or (34) it follows that

$$\mathcal{D}^{-1}(\mathbf{r}', \mathbf{r}, \omega = 0) = \mathcal{D}^{*-1}(\mathbf{r}', \mathbf{r}, \omega = 0) - V(\mathbf{r}' - \mathbf{r}). \tag{36}$$

This shows the well-known fact that in the formulation which employs the density as variable there is no need to solve the Dyson equation (35) or (34) in order to obtain the second-order coefficient. Based on Eq. (36), it suggests itself to split up the term quadratic in the density and to write the term involving the classical Coulomb interaction separately [1,53]

$$F_\beta[n] = \frac{1}{2} \int d^3 r d^3 r' n(\mathbf{r}) V(\mathbf{r} - \mathbf{r}') n(\mathbf{r}') + G_\beta[n]. \tag{37}$$

The first term on the right-hand side of Eq. (37) is the only one that is common to all expansions of $F_\beta[n]$ irrespective of the starting point n_0 .

To gain further insight into the formal structure of the expansion of $F_\beta[n]$ we consider two typical diagrams contributing to $\frac{\delta^3}{\delta v'(\mathbf{r}_3) \delta v'(\mathbf{r}_2) \delta v'(\mathbf{r}_1)} \Omega[-\beta v'](\beta)$. In the first diagram in Fig. 6 we may separate off the part carrying the label $\mathbf{r}_1$ by cutting through the vertex line indicated by the arrow. The same is true for the parts carrying $\mathbf{r}_2$ and $\mathbf{r}_3$. By convention we always separate off the largest possible part, if there are several possibilities, as it is the case for the $\mathbf{r}_3$ label in the second diagram. It may also happen that it is not possible to isolate the label-carrying part by cutting through a single vertex line, as it is the case for the label $\mathbf{r}_1$ in the second diagram. The sum of all subdiagrams which can be separated off from the diagrams contributing to the third-order term and leaving behind a common subdiagram is equal to

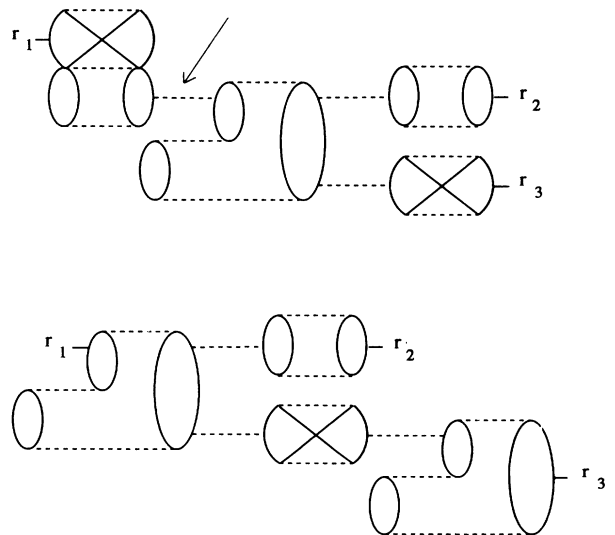


FIG. 6. Feynman diagrams contributing to the third-order term $\frac{\delta^3}{\delta v'(\mathbf{r}_3) \delta v'(\mathbf{r}_2) \delta v'(\mathbf{r}_1)} \Omega[-\beta v'](\beta)$. The propagator lines are not oriented since their orientation is irrelevant in the present context.

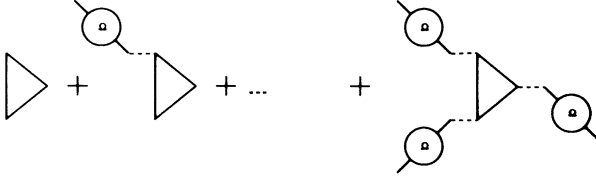


FIG. 7. Structure of the third-order term $\frac{\delta^3}{\delta v'(\mathbf{r}_3)\delta v'(\mathbf{r}_2)\delta v'(\mathbf{r}_1)}\Omega[-\beta v'](\beta)$, in terms of skeleton diagrams. The triangle contains all possible diagrams with three indices, which do not fall into unconnected parts by cutting through a single vertex line.

$\frac{\delta^2}{\delta v'(\mathbf{r}_3)\delta v'(\mathbf{r}_1)}\Omega[-\beta v'](\beta)$. Therefore, the third derivative of $\Omega[-\beta v'](\beta)$ with respect to v' has a structure shown in Fig. 7. It is now easy to verify that the $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ -kernels appearing in the $\frac{\delta^3}{\delta v'(\mathbf{r}_3)\delta v'(\mathbf{r}_2)\delta v'(\mathbf{r}_1)}\Omega[-\beta v'](\beta)$ are canceled by the black dots in the expansion of $F_\beta[n]$. This property of the third-order term, namely, that it cannot be divided into label-carrying parts by separating sub-diagrams contributing to $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$, will be called $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ irreducibility. In fourth order of the density an additional new type of $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ reducible diagrams appears, namely, one which falls into parts that carry two indices if the largest possible diagrams contributing to $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ are cut out. An example is given by the terms contributing to the first skeleton diagram in Fig. 8. This new type of $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ reducible diagram is canceled by a corresponding term of the first summand of the fourth-order term of Eq. (11). This result coincides with the findings of Osaka [14]. Proceeding in the indicated way, order by order with the analysis of the expansion (the general structure of the m th-order term is given in the Appendix), it can be verified that no $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ reducible contributions remain in the expansion. The $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ irreducibility of Eq. (11) represents the analog to the one-line irreducibility of the Legendre transformed generating functional in field theory (see [59], p. 114). This result shows that the expansion of $F_\beta[n]$ does not contain terms which account for

density fluctuations $[\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ describes the static density fluctuations]. The density enters as a classical variable. Collecting only the leading terms in each order, we may look at Eq. (11) as a perturbation expansion in

the local potential $\bullet \bigcirc \Delta n$. The remaining terms account for the relaxation of the reference medium under the influence of this potential.

In addition to the expansion of $F_\beta[n]$ discussed above, further explicit representations of quantities, which are of interest in density-functional theory, can now be generated. The Kohn-Sham potential [2], for instance, will be obtained as the derivative of $G_\beta[n]$ with respect to Δn after deleting all terms which are of zeroth order in the electron-electron interaction. The latter contributions give the kinetic energy of the noninteracting electrons.

Static observables of the electron system are obtained by calculating the derivative of the grand canonical potential with respect to a parameter λ . λ denotes the coupling strength with which a suitable operator $\hat{O}$ is added to the Hamiltonian. The grand canonical potential, its Legendre transformed functional $F_\beta[n]$, and the electron density n become a function of λ

$$\Omega[-\beta v'](\beta)(\lambda) = F_\beta[n_0(\lambda)](\lambda) + \int d^3r n_0(\mathbf{r}, \lambda) v'(\mathbf{r}). \quad (38)$$

By using the fact that $\Omega[-\beta v'](\beta)(\lambda)$ is stationary for variations of n , we obtain, for the derivative of the grand canonical potential with respect to λ ,

$$\frac{d}{d\lambda} \Omega[-\beta v'](\beta)(\lambda) = \langle \hat{O} \rangle = \frac{\partial}{\partial \lambda} F_\beta[n_0(\lambda)](\lambda). \quad (39)$$

The formal calculation of $F_\beta^\lambda[n_0(\lambda)](\lambda)$ starting with Eq. (11) causes no difficulties. An analogous formula to Eq. 39 for property density functionals within the Kohn-Sham formalism has been derived by Bauer [60] and G rling, Levy, and Perdew [61].

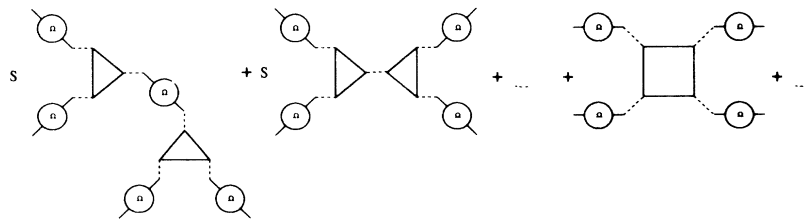


FIG. 8. Skeleton diagrams of the fourth-order term $\frac{\delta^4}{\delta v'(\mathbf{r}_4)\delta v'(\mathbf{r}_3)\delta v'(\mathbf{r}_2)\delta v'(\mathbf{r}_1)}\Omega[-\beta v'](\beta)$. The square, like the triangle, does not contain any diagrams separable into unconnected parts by cutting through a single vertex line. The first contribution falls into parts containing two external legs by cutting out $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$. Note that the frequency ω of the inner $\mathcal{D}(\mathbf{r}', \mathbf{r}, \omega = 0)$ matrix in the first term is fixed to be zero by frequency conservation. The S in front of the diagrams indicates that the skeleton diagrams are to be symmetrized with respect to permutations of the labeled legs. In the above example this leads to two additional diagrams respectively, since there are three possibilities to group the four labeled legs into pairs.

The case in which the perturbing potential is given by an external local potential and the properties to be calculated are the static response functions merits special attention. The defining equations of the Legendre transformation exhibit a perfect symmetry between $\Omega[-\beta v'](\beta)$ and $F_\beta[n]$ apart from the sign, which is a matter of convention. As a consequence the expansion of $\Omega[-\beta v'](\beta)$ in terms of the perturbing potential is given by the same graphic expression as the expansion of $F_\beta[n]$ with respect to the density, where the derivatives of $\Omega[-\beta v'](\beta)$ in the expansion coefficients are to be replaced by the derivatives of $-F_\beta[n]$.

IV. CONCLUSION

Based on the construction scheme for the Taylor series of $F[n]$, explicit expressions have been obtained for the expansions of the density functionals of HK and Mermin as well as for property density functionals. Furthermore, we were able to discuss the relation between density functional theory and the field-theoretical approach to the many-particle problem. It turns out that the inverse of the second-order polarization propagator of the reference many-particle system plays the role of the zeroth-order propagator. Like the higher-order polarization propagators, as the interaction vertices of the diagrammatic expansion, they depend on the chosen reference point. In order to remove this dependency, resummations of this expansion have to be done. An expansion of the Kohn-Sham potential has already been described in Sec. III. Alternatively we can obtain this potential by an iterative resummation. Let n denote a given electron density of an interacting electron system. We make an expansion of the potential v , which gives rise to n if applied to a noninteracting system (i.e., the reference system is a noninteracting system). This expansion is made around a certain reference point v_0 and will be truncated after the linear contribution in Δn . The approximate potential resulting from the latter step will be used as a new reference point in a further expansion of v up to the linear term in the density. By formulating the procedure described in terms of diagrams, it can readily be verified that in each iteration of this procedure a particular class of diagrams of the expansion of v is accounted for, i.e., the procedure converges to the exact v . The Kohn-Sham potential can be obtained from v by subtracting the external potential and the Coulomb potential caused by the electron density. The described approach is identical to the method for the evaluation of the Kohn-Sham potential proposed by Görling [62]. Görling and the present author have given a number of examples for the calculation of Kohn-Sham potentials for atomic systems [63].

As an example for the explicit construction of density functionals, we have investigated the kinetic-energy functional of a one-dimensional noninteracting electron system. Starting from a homogeneous electron density as reference point, the exact expansion of the kinetic-energy functional up to fourth order has been evaluated for various particle numbers. Important and well understood features to be described by such a functional are,

for example, the endeavor of the electrons to form electron pairs, which is closely related to the derivative discontinuity. But obviously, the electron pairs interact with each other and our analysis enables a quantitative understanding of this effect. One point, for instance, which has not been recognized so far is the tendency towards the formation of electron quadruples. This effect is again a consequence of the derivative discontinuity or vice versa. Tendencies towards the formation of hexuples, etc. are likely to appear in the higher-order terms of the expansion. The analogous behavior for bosonic and electronic systems in second order of rapidly oscillating densities is shown to have a generalization in any order.

In general, it turns out that the kinetic-energy functional in one dimension has a highly complicated and singular structure in the reciprocal space, which implies that it is totally non-local in real space so that it cannot be described by *Ansätze* involving a finite number of gradient terms. In particular this is true if the particle number tends to infinity. Moreover, the infinite system shows a degenerate ground state, which completely disqualifies it as a reference point for any series expansion. The situation in three dimensions may be less serious since the Lindhard density response function of the noninteracting system in three dimensions behaves more regularly than the one-dimensional counterpart. However, one should be aware of the fact that the three-dimensional kinetic-energy functional reduces to the one-dimensional kinetic-energy functional if the extensions of the electron system becomes small in two spatial directions.

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APPENDIX: LEGENDRE TRANSFORMATION

In order to explicitly perform a Legendre transformation we consider an example of a functional $\omega = \omega[q]$, $q = q(\mathbf{r})$ and demonstrate the essential steps which lead to an expansion of the Legendre-transformed functional $f[p]$. The system of equations that defines the relation between $\omega[q]$ and $f[p]$ is given by

$$-f[p] + \omega[q] = \int d^3r q(\mathbf{r}) p(\mathbf{r}), \quad (\text{A1})$$

$$p(\mathbf{r}) = \frac{\delta}{\delta q(\mathbf{r})} \omega[q], \quad (\text{A2})$$

$$q(\mathbf{r}) = -\frac{\delta}{\delta p(\mathbf{r})} f[p]. \quad (\text{A3})$$

Apart from the signs which are a matter of convention, the equations exhibit the symmetry between $f[p]$ and $\omega[q]$. $q(\mathbf{r})$ is usually eliminated from Eq. (A1) by using Eq. (A2) to obtain an expression for $f[p]$ in terms of $p(\mathbf{r})$. This elimination can be achieved by inverting the expression Eq. (A2). The inversion is locally, i.e., close to a point q_0 , possible if and only if $\frac{\delta p_0(\mathbf{r})}{\delta q(\mathbf{r}')} = \frac{\delta^2}{\delta q(\mathbf{r}')\delta q(\mathbf{r})}\omega[q_0]$ does not become singular, or in other words, if there locally exists a one-to-one correspondence between q and p .

In order to construct $f[p]$ we make use of the Taylor-series expansions of $\omega[q]$ and $f[p]$ around $\omega[q_0]$ and $f[p_0]$, respectively. We assume that $p_0(\mathbf{r}) = \frac{\delta}{\delta q(\mathbf{r})}\omega[q_0]$ and obtain

$$\begin{aligned} \omega[q_0 + \Delta q] &= \omega[q_0] + \int d^3r p_0(\mathbf{r}) \Delta q(\mathbf{r}) \\ &+ \frac{1}{2!} \int d^3r d^3r' \Delta q(\mathbf{r}) \left(\frac{\delta^2 \omega[q_0]}{\delta q(\mathbf{r}') \delta q(\mathbf{r})} \right) \\ &\times \Delta q(\mathbf{r}') + \frac{1}{3!} \dots, \end{aligned} \quad (\text{A4})$$

$$\begin{aligned} f[p_0 + \Delta p] &= f[p_0] - \int d^3r q_0(\mathbf{r}) \Delta p(\mathbf{r}) \\ &+ \frac{1}{2!} \int d^3r d^3r' \Delta p(\mathbf{r}') \\ &\times \left(\frac{\delta^2 f[p_0]}{\delta p(\mathbf{r}') \delta p(\mathbf{r})} \right) \Delta p(\mathbf{r}) + \frac{1}{3!} \dots. \end{aligned} \quad (\text{A5})$$

To calculate the expansion coefficients

$$\left(\frac{\delta^2 f[p_0]}{\delta p(\mathbf{r}_2) \delta p(\mathbf{r}_1)} \right)$$

we proceed as in the Introduction and start with the equation

$$\frac{\delta}{\delta p(\mathbf{r}_2)} p(\mathbf{r}) = \frac{\delta}{\delta p(\mathbf{r}_2)} \frac{\delta}{\delta q(\mathbf{r})} \omega[q]. \quad (\text{A6})$$

We employ the chain rule, i.e.,

$$\frac{\delta}{\delta p(\mathbf{r}_2)} = \int d^3r' \frac{\delta q(\mathbf{r}')}{\delta p(\mathbf{r}_2)} \frac{\delta}{\delta q(\mathbf{r}')}, \quad (\text{A7})$$

to calculate the rhs of Eq. (A6) and end up with

$$\delta(\mathbf{r}_2 - \mathbf{r}) = - \int d^3r' \frac{\delta^2 f[p]}{\delta p(\mathbf{r}_2) \delta p(\mathbf{r}')} \frac{\delta^2 \omega[q]}{\delta q(\mathbf{r}') \delta q(\mathbf{r})} \quad (\text{A8})$$

or

$$\frac{\delta^2 f[p]}{\delta p(\mathbf{r}_2) \delta p(\mathbf{r}_1)} = - \left(\frac{\delta^2 \omega[q]}{\delta q(\mathbf{r}_2) \delta q(\mathbf{r}_1)} \right)^{-1}. \quad (\text{A9})$$

In order to calculate higher derivatives of the last equation with respect to $p(\mathbf{r})$ we introduce a graphical notation ([64], pp. 115)

$$\frac{\delta}{\delta p(\mathbf{r})} f[p] = \text{---} \bigcirc \text{---} f \quad (\text{A10})$$

Each further differentiation with respect to $q(\mathbf{r})$ increases the number of legs by one

$$\frac{\delta^n}{\delta p(\mathbf{r}_n) \dots \delta p(\mathbf{r}_1)} f[p] = \text{---} \bigcirc \text{---} f \begin{matrix} \text{etc.} \\ \uparrow \\ n \\ \uparrow \\ 2 \\ \uparrow \\ 1 \end{matrix} \quad (\text{A11})$$

For the derivative of $\omega[q]$ with respect to $q(\mathbf{r})$ we use the symbol

$$\frac{\delta}{\delta q(\mathbf{r})} \omega[q] = \text{---} \bigcirc \text{---} \omega \quad (\text{A12})$$

Higher derivatives of $\omega[q]$ are represented analogous to Eq. A11.

By using the chain rule A7 and taking successively higher derivatives of

$$\text{---} \bigcirc \text{---} f = - \text{---} \bigcirc \text{---} \omega^{-1}, \quad (\text{A13})$$

we obtain

$$\text{---} \bigcirc \text{---} f = - \begin{matrix} \text{---} \bigcirc \text{---} f \\ | \\ \bigcirc \text{---} \omega \text{---} \bigcirc \text{---} f \\ | \\ \bigcirc \text{---} f \end{matrix}, \quad (\text{A14})$$

$$\begin{aligned}
 f[p] = f[p_0] &+ \text{diagram 1} - \frac{1}{2!} \text{diagram 2} + \frac{1}{3!} \text{diagram 3} \\
 &- \frac{3}{4!} \text{diagram 4} + \frac{1}{4!} \text{diagram 5} + \dots,
 \end{aligned}
 \tag{A17}$$

where denotes the inverse of the kernel $\frac{\delta^2 \omega[q]}{\delta q(\mathbf{r}') \delta q(\mathbf{r})}$.

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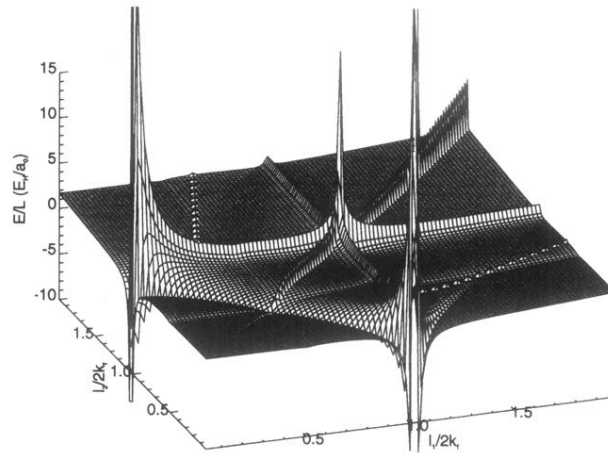


FIG. 4. The term $F_4[n]/L$ for a system with 102 electrons and a cycle length of 102.

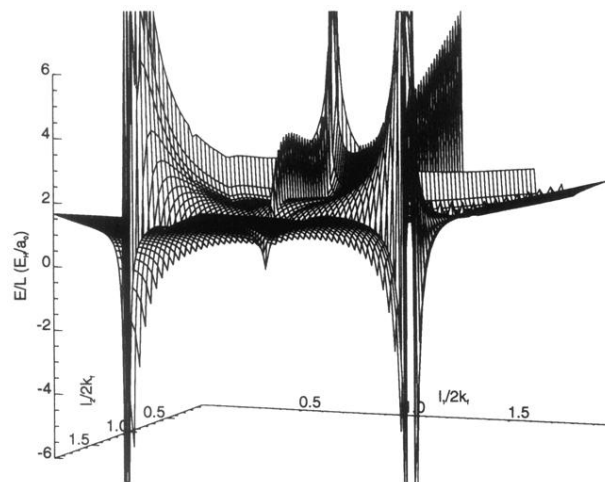


FIG. 5. The graph of Fig. 4, shown from below.